SOFTWARE LAB-04

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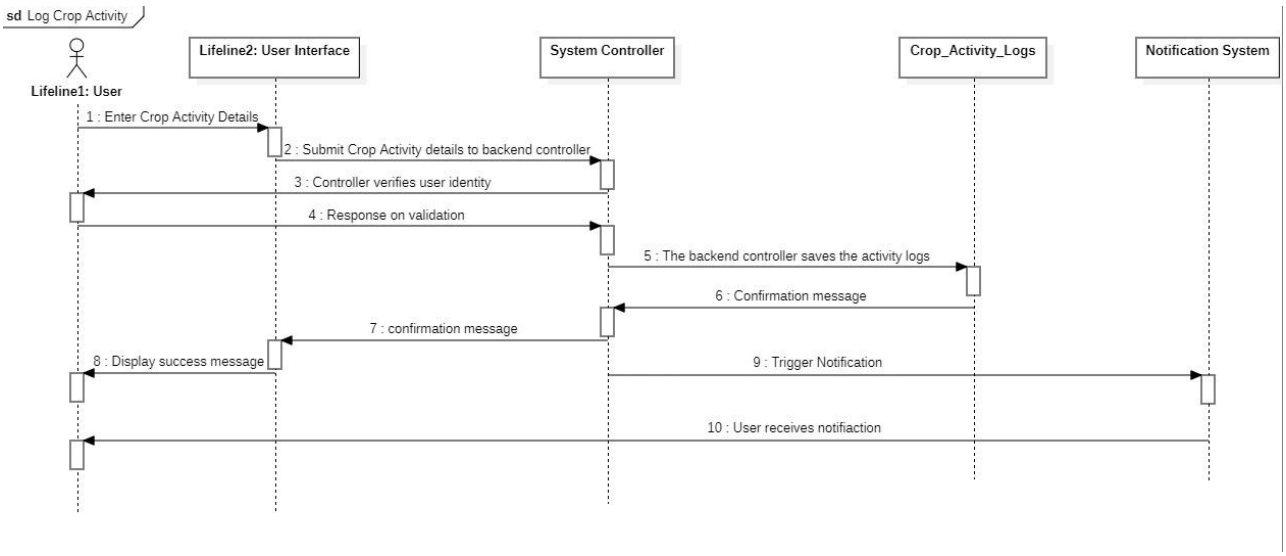
Sequence Diagram



This sequence diagram represents the step-by-step process for logging crop activity within a system, showing the interaction between the User, User Interface (UI), System Controller, Crop Activity Logs, and Notification System.

Participants:

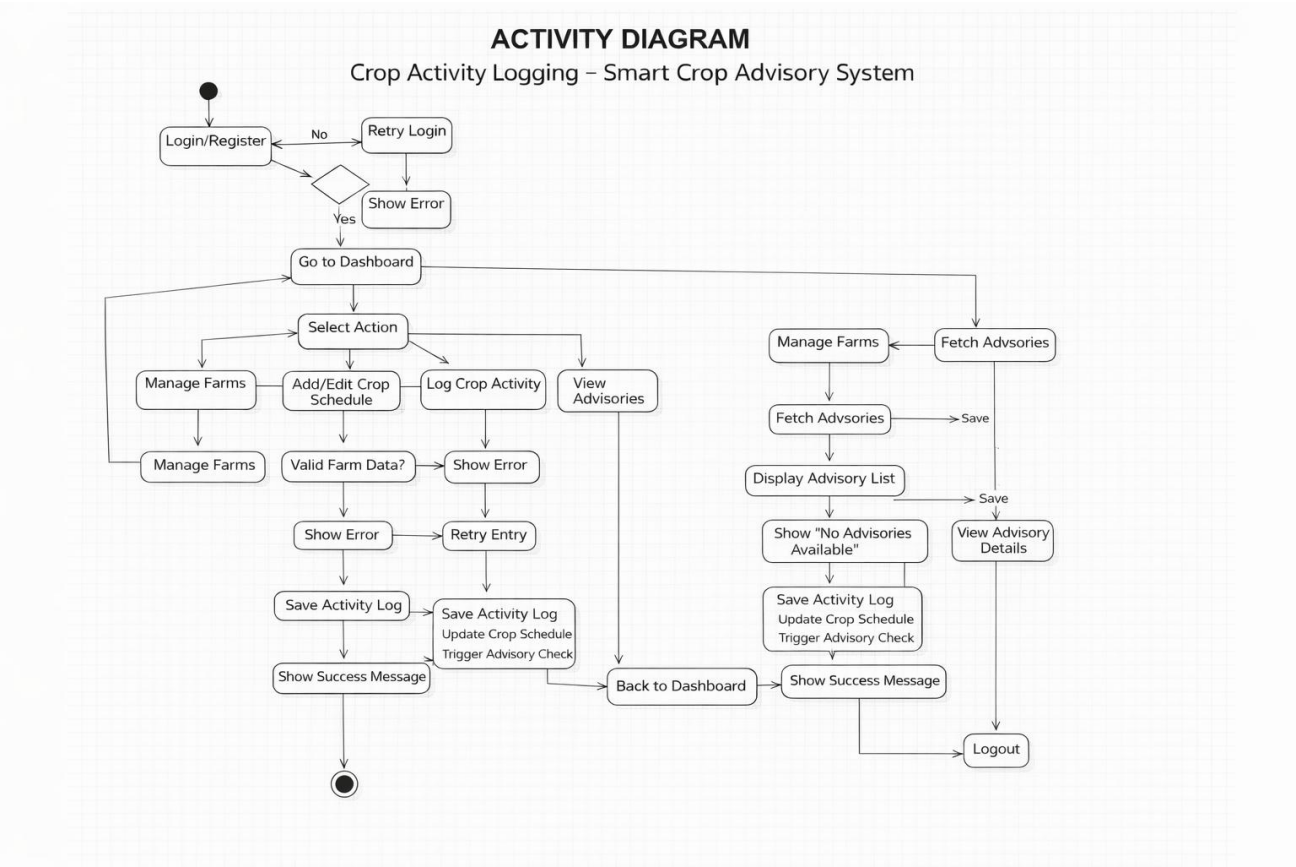
- **User:** The individual initiating the crop activity logging.
- **User Interface (UI):** The front-end system interface through which the user interacts.
- **System Controller:** Backend component that manages validation, data processing, and coordination.
- **Crop Activity Logs:** The data storage system where crop activity details are saved.
- **Notification System:** Responsible for sending notifications to the user.

Flow of Activities:

1. User Input: The process starts with the User entering crop activity details through the User Interface (Step 1: Enter Crop Activity Details).
2. Submission to Backend: The User Interface sends the entered crop activity details to the System Controller for processing (Step 2: Submit Crop Activity details to backend controller).
3. User Validation: The System Controller verifies the identity of the user to ensure they are authorized to perform this action (Step 3: Controller verifies user identity).
4. Validation Response: The validation result is sent back from the System Controller to the User Interface (Step 4: Response on validation).
5. Saving Activity Logs: Once the user is validated, the System Controller saves the crop activity logs to the Crop Activity Logs system (Step 5: The backend controller saves the activity logs).
6. Confirmation Message: The Crop Activity Logs sends a confirmation message back to the System Controller indicating successful saving of data (Step 6: Confirmation message).
7. Notify UI: The System Controller sends a confirmation message to the User Interface to inform that the crop activity has been successfully logged (Step 7: Confirmation message).

8. Display Success: The User Interface displays a success message to the User confirming that the crop activity has been recorded (Step 8: Display success message).
9. Trigger Notification: The System Controller triggers a notification to the Notification System about the newly logged crop activity (Step 9: Trigger Notification).
10. User Notification: Finally, the Notification System sends a notification to the User, completing the process (Step 10: User receives notification).

Activity Diagram:



- The process starts with Register/Login**.**
- If login credentials are invalid, the system shows an error and allows the user to retry.
- If login is successful, the user is redirected to the Dashboard**.**
- From the dashboard, the user selects an action such as:
  - Manage Farms
  - Add/Edit Crop Schedule
  - Log Crop Activity
  - View Advisories
- If Log Crop Activity is selected:
  - The user enters activity details.
  - The system validates the input.
  - If invalid, an error is shown and the user retries.
  - If valid, the system saves the activity log, updates the crop schedule, and may trigger advisory checks.
  - A success message is displayed.
  - The user is redirected back to the dashboard.
- If View Advisories is selected:
  - The system fetches and displays advisory details.
  - The user can view specific advisory information.
  - The system returns to the dashboard.
- The process ends when the user selects Logout**.**